

GBM vs. XGBoost

Step-by-Step Mathematical Comparison

Every step: what changes, and why it matters

Step	Gradient Boosting (GBM)	XGBoost — What Changes?	Benefit over GBM
1. Initial Prediction	Regression: mean of target values. $F_0 = \text{mean}(y)$	Same idea. Starts with the base prediction that minimises the loss.	No major difference.
2. Compute Error Signal	For MSE, computes residuals (which are also negative gradients): $r_i = y_i - \hat{y}_i$	Computes gradient g_i and Hessian h_i for every sample: $g_i = \partial_{\hat{y}} L(y_i, \hat{y}_i), \quad h_i = \partial_{\hat{y}}^2 L(y_i, \hat{y}_i)$	Knows both the direction and curvature of the loss. More accurate optimisation.
3. What is Passed to the Tree?	Residuals / negative gradients only.	Gradient–Hessian pairs (g_i, h_i) for each sample.	Tree has richer information about the loss surface.
4. Build Tree / Find Splits	Split chosen by reduction in impurity (typically MSE reduction).	Split chosen using the Gain Formula : $\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right]$	Mathematically optimal, regularisation-aware splits.
5. Role of γ (Gamma)	Not present.	Used during split evaluation. Every split must produce enough gain to overcome γ : $\text{Split accepted} \iff \text{Gain} > \gamma$	Prevents unnecessary leaves and directly controls tree complexity.
6. Split Acceptance	If impurity decreases, the split is usually accepted.	Split accepted only if $\text{Gain} > \gamma$ after the γ penalty.	Better generalisation; less overfitting.
7. Leaf Formation	Samples with similar residuals are grouped into leaves.	Samples with similar gradient–Hessian statistics $(\sum g_i, \sum h_i)$ are grouped into leaves.	More statistically informed grouping of samples.

Summary in one line per symbol: g_i = direction of loss | h_i = curvature of loss | λ = shrinks leaf weights (L_2) | α = sparsifies leaf weights (L_1) | γ = minimum gain to allow a split | $w_j^* = -G_j/(H_j + \lambda)$ = optimal leaf score